

Majoring & Minorin

5–7 minutes

Major in Computer Science (BA): 10 courses

- CSC54-184 Computer Science I
- CSC54-284 Computer Science II
- CSC54-384 Discrete Mathematics
- CSC54-394 Computer Organization
- CSC54-454 Algorithms
- CSC54-474 Programming Languages
- CSC54-644 Computer Systems
- CSC54-894 Senior Seminar in Software Engineering (Capstone, to be taken in the Spring of senior year)
- Two additional approved upper-level courses in Computer Science, normally begun no later than junior year

Required supporting course in the Computer Science major (BA): 1 course

- MAT52-164 Modern Calculus I, normally to be completed no later than the sophomore year

Major in Computer Science (BS): 10 courses

- CSC54-184 Computer Science I
- CSC54-284 Computer Science II

- CSC54-384 Discrete Mathematics
- CSC54-394 Computer Organization
- CSC54-454 Algorithms
- CSC54-474 Programming Languages
- CSC54-644 Computer Systems
- CSC54-894 Senior Seminar in Software Engineering (Capstone, to be taken in the Spring of senior year)
- Two additional approved upper-level courses in Computer Science, normally begun no later than junior year

Required supporting courses in the Computer Science major (BS): 7 courses

- BIO50-173/171 Molecular and Cellular Foundations of Biology/lab,
or
BIO50-183/181 Evolutionary and Ecological Foundations of Biology/lab
- CHE51-103/101 General Chemistry I/Chemical Methods and Techniques Laboratory I
- MAT52-164 Modern Calculus I
- MAT52-264 Modern Calculus II
- MAT52-674 Linear Algebra
- PHY53-154 Fundamentals of Physics
- One course from:

BIO50-183/181 Evolutionary and Ecological Foundations of Biology/lab, and
BIO50-173/171 Molecular and Cellular Foundations of Biology/lab
(counts as one course here since one is required above)

CHE51-203/201 General Chemistry II/Chemical Methods and

Techniques Laboratory II

MAT52-114 Introduction to Statistics

PHY53-164 Fundamentals of Physics II

PHY53-404 Electronics

PHY53-454 Math Methods in Physical Science

- Additional approved upper-level elective in Computer Science or Mathematics

Minor in Computer Science: 5 courses

- CSC54-184 Computer Science I
- CSC54-284 Computer Science II
- One course from:
 - CSC54-394 Computer Organization
 - CSC54-454 Algorithms
- Two additional approved upper-level courses in Computer Science

Major in Computational Mathematics (BA): 14 courses

- CSC54-184 Computer Science I
- CSC54-284 Computer Science II
- CSC54-384 Discrete Mathematics
- CSC54-394 Computer Organization
- CSC54-454 Algorithms
- MAT52-164 Modern Calculus I
- MAT52-264 Modern Calculus II
- MAT52-364 Modern Calculus III
- MAT52-674 Linear Algebra
- MAT52-754 Differential Equations

- Two courses from:
CSC54-414 Operations Research
CSC54-514 Database Management
MAT52-524 Introduction to Numerical Analysis
MAT52-574 Probability and Mathematical Statistics
- One additional approved upper-level course, in Mathematics or Computer Science or PHY53-454 Math Methods in Physical Science
- MAT52-894 Senior Seminar in Mathematics (Capstone, to be taken in the Fall of senior year) or CSC54-894 Senior Seminar in Software Engineering (Capstone, to be taken in the Spring of senior year)

Major in Computational Mathematics (BS): 15 courses

- CSC54-184 Computer Science I
- CSC54-284 Computer Science II
- CSC54-384 Discrete Mathematics
- CSC54-394 Computer Organization
- CSC54-454 Algorithms
- MAT52-164 Modern Calculus I
- MAT52-264 Modern Calculus II
- MAT52-364 Modern Calculus III
- MAT52-674 Linear Algebra
- MAT52-754 Differential Equations
- Two courses from:
CSC54-414 Operations Research
CSC54-514 Database Management

MAT52-524 Introduction to Numerical Analysis

MAT52-574 Probability and Mathematical Statistics

- Two additional approved upper-level courses, in Mathematics or Computer Science or PHY53-454 Math Methods in Physical Science
- MAT52-894 Senior Seminar in Mathematics (Capstone, to be taken in the Fall of senior year) or CSC54-894 Senior Seminar in Software Engineering (Capstone, to be taken in the Spring of senior year)

Required supporting courses in the Computational Mathematics major (BS): 4 courses

- BIO50-173/171 Molecular and Cellular Foundations of Biology/lab,
or
BIO50-183/181 Evolutionary and Ecological Foundations of Biology/lab
- CHE51-103/101 General Chemistry I/Chemical Methods and Techniques Laboratory I
- PHY53-154 Fundamentals of Physics I
- One course from:

BIO50-183/181 Evolutionary and Ecological Foundations of Biology/lab, and

BIO50-173/171 Molecular and Cellular Foundations of Biology/lab
(counts as one course here since one is required above)

CHE51-203/201 General Chemistry II/Chemical Methods and Techniques Laboratory II

PHY53-164 Fundamentals of Physics II

PHY53-404 Electronics

See the Education Department for information regarding teacher

certification in Mathematics and Computer Science.

Mathematics

See [Mathematics](#) for information regarding a Mathematics major.

Data Science

See [Data Science](#) for information regarding the 18 credits required for a Data Science minor

Education

See the [Education Department](#) for information regarding the 24 credits required for secondary or all-level certification in mathematics or computer science.

Engineering

See the Dual Degree, [Applied Physics](#) regarding the 16 mathematics credits required for engineering.